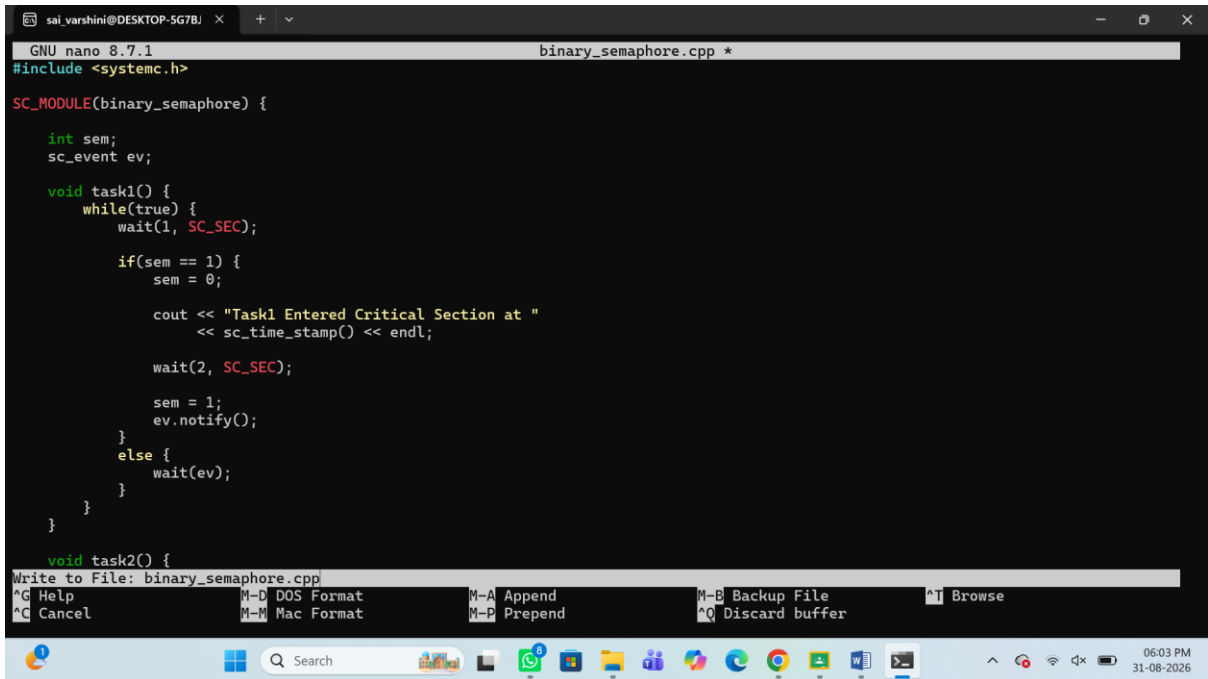


Exp No: 27

BINARY SEMAPHORE IN SYSTEMC



```
GNU nano 8.7.1 binary_semaphore.cpp *
#include <systemc.h>

SC_MODULE(binary_semaphore) {

    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);

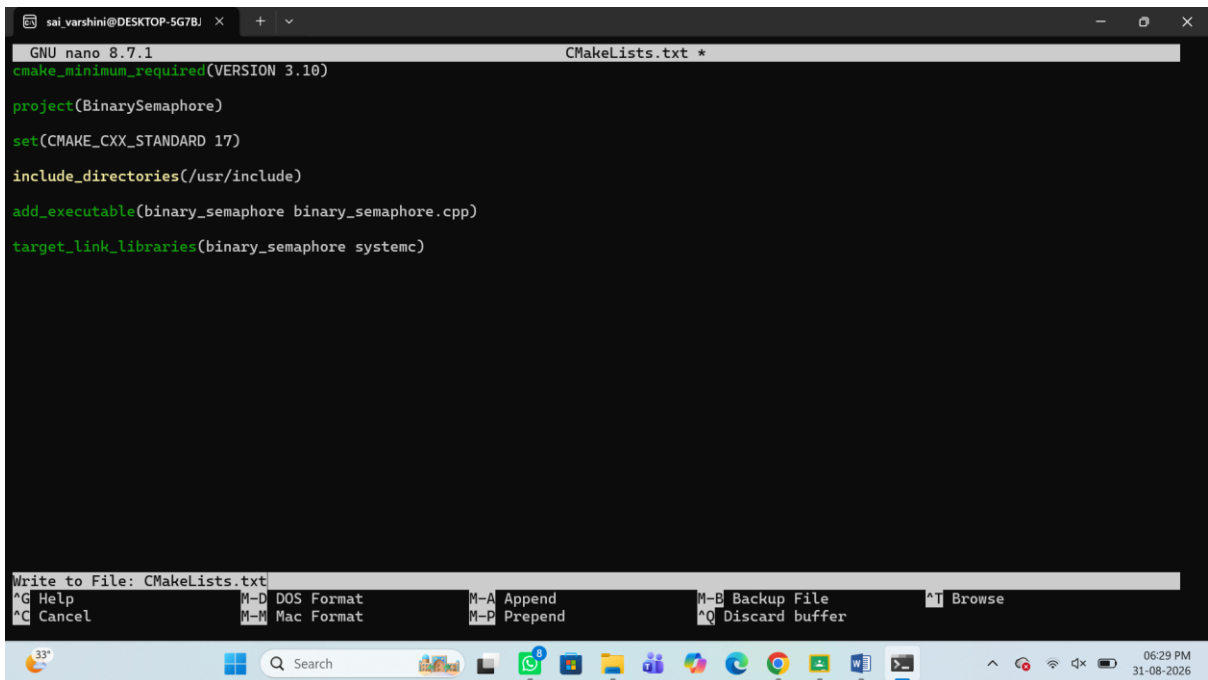
            if(sem == 1) {
                sem = 0;

                cout << "Task1 Entered Critical Section at "
                     << sc_time_stamp() << endl;

                wait(2, SC_SEC);

                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
```



```
GNU nano 8.7.1 CMakeLists.txt *
cmake_minimum_required(VERSION 3.10)

project(BinarySemaphore)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(binary_semaphore binary_semaphore.cpp)

target_link_libraries(binary_semaphore systemc)
```

OUTPUT:

```
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27$ ls
CMakeLists.txt  binary_semaphore.cpp
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27$ mkdir build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27$ cd build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27/build$ cmake ..
-- The C compiler identification is GNU 15.2.0
-- The CXX compiler identification is GNU 15.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (5.3s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sai_varshini/systemc_test/exp27/build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27/build$ make
[ 50%] Building CXX object CMakeFiles/binary_semaphore.dir/binary_semaphore.cpp.o
[100%] Linking CXX executable binary_semaphore
[100%] Built target binary_semaphore
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27/build$ ./binary_semaphore

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp27/build$ |
```